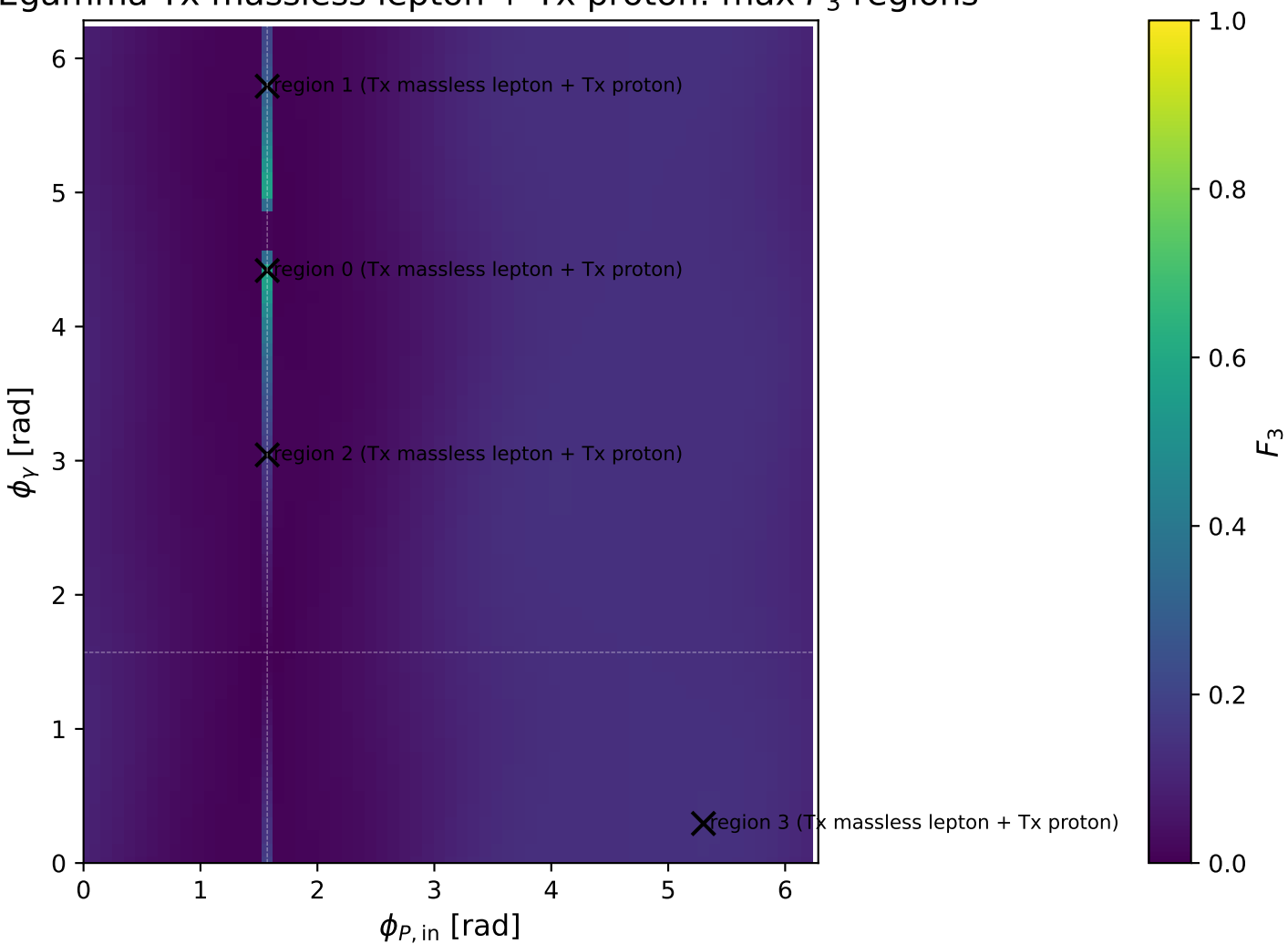
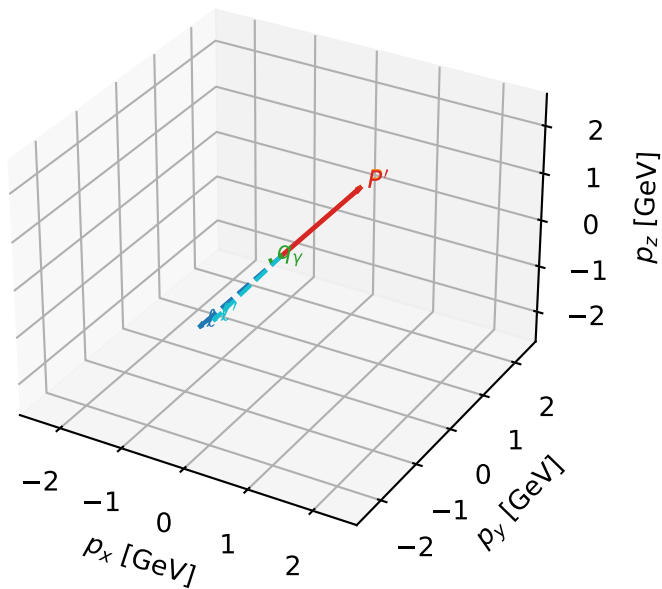


# Egamma Tx massless lepton + Tx proton: max $F_3$ regions



# Tx massless lepton + Tx proton characteristic max $F_3$ configuration

3D momenta



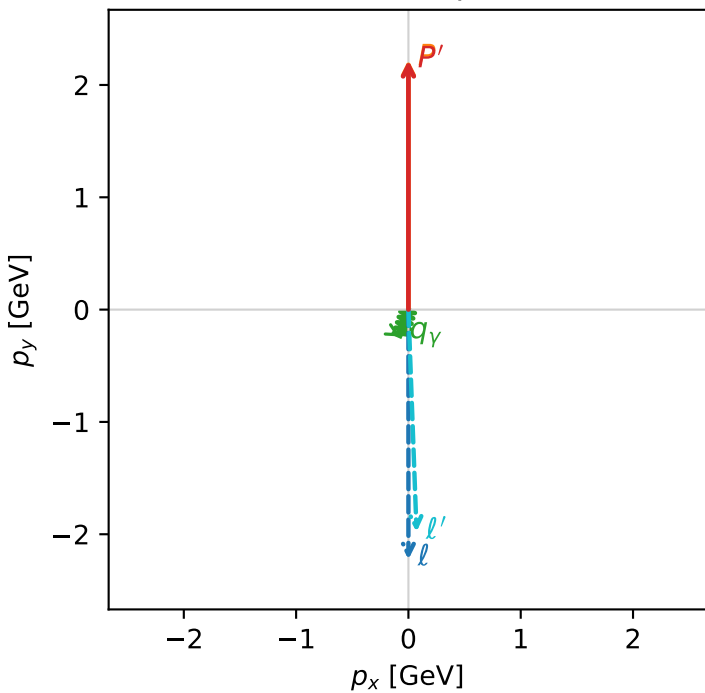
f3\_low\_egamma\_txtx\_region\_0  $F_3=0.572473$   
 region: low\_Egamma\_TxTx\_region\_0  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.21812$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=4.41786$   
 $Q^2=0.00591804$ ,  $x_B=0.00260549$ ,  $t=-5.99798e-06$ ,  $W^2=3.1453$ ,  $y=0.110068$   
 $\theta(\ell', \gamma)=18.9753$  deg

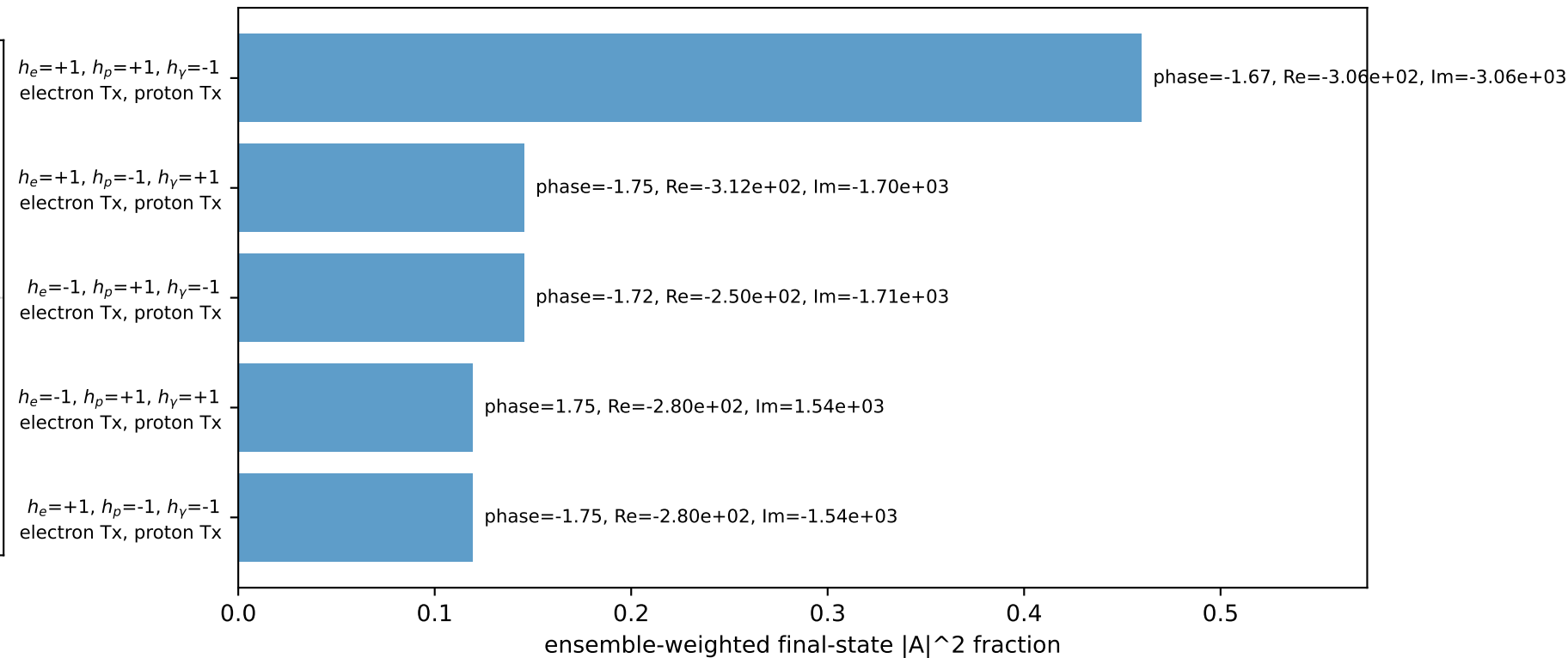
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 1.9802$   $p_x= 0.0726$   $p_y= -1.9789$   $p_z= 0.0000$   
 $P'$   $E= 2.4083$   $p_x= 0.0000$   $p_y= 2.2181$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= -0.0726$   $p_y= -0.2392$   $p_z= 0.0000$

Transverse plane

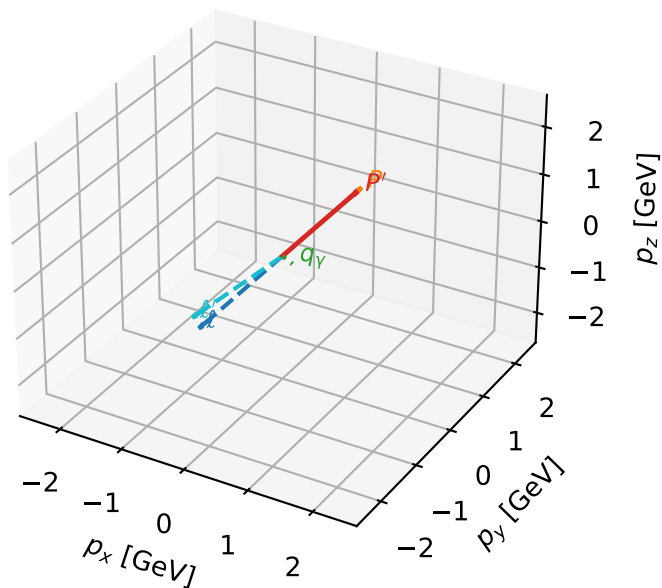


Leading initial-ensemble/final-state components (N=5, retained=98.9%)



# Tx massless lepton + Tx proton characteristic max $F_3$ configuration

3D momenta



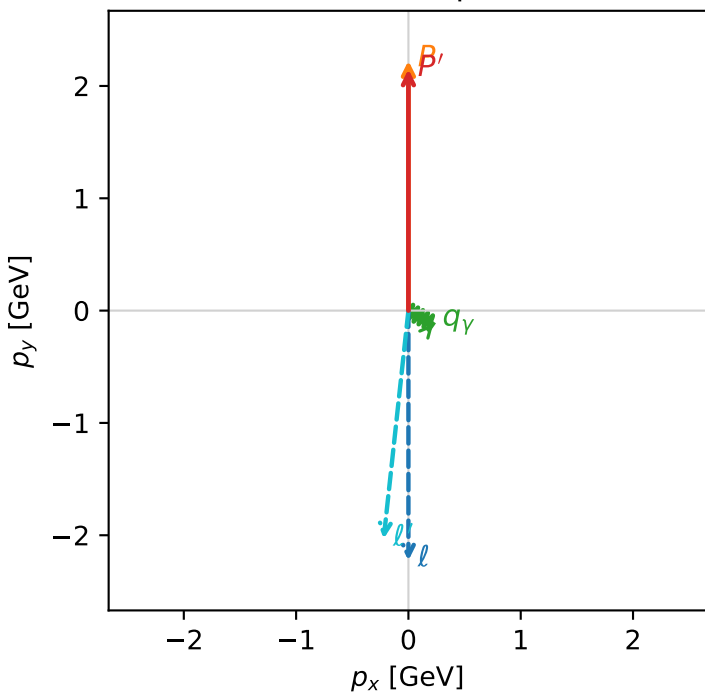
f3\_low\_egamma\_txtx\_region\_1 F\_3=0.287339  
 region: low\_Egamma\_TxTx\_region\_1  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.14934$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=5.79231$   
 $Q^2=0.0530724$ ,  $x_B=0.0306384$ ,  $t=-0.000876114$ ,  $W^2=2.55899$ ,  $y=0.0839415$   
 $\theta(\ell', \gamma)=68.0691$  deg

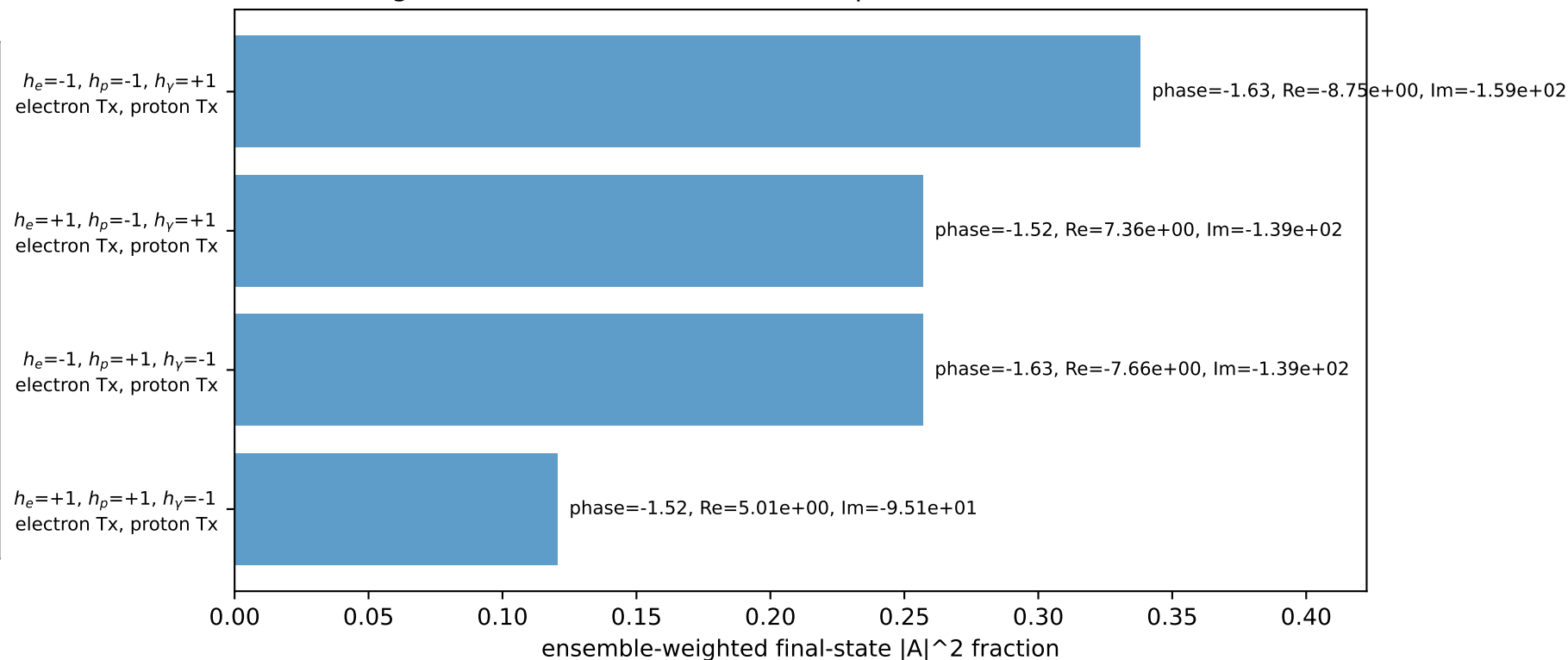
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.0434$   $p_x= -0.2205$   $p_y= -2.0315$   $p_z= 0.0000$   
 $P'$   $E= 2.3451$   $p_x= 0.0000$   $p_y= 2.1493$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.2205$   $p_y= -0.1178$   $p_z= 0.0000$

Transverse plane

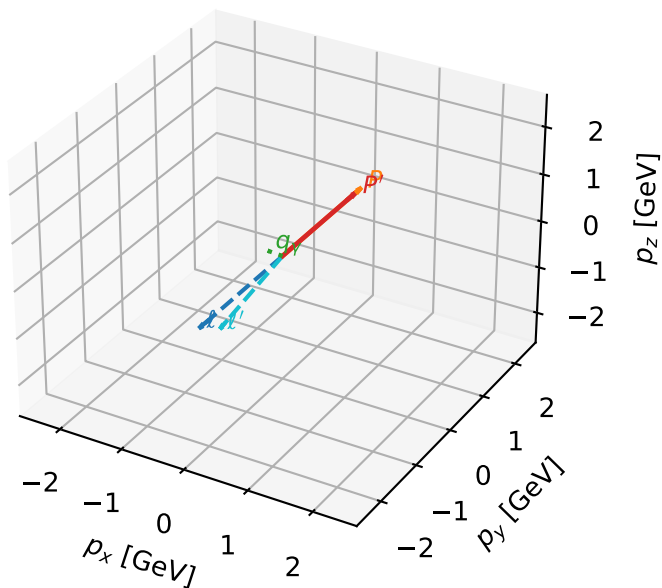


Leading initial-ensemble/final-state components (N=4, retained=97.3%)



# Tx massless lepton + Tx proton characteristic max $F_3$ configuration

3D momenta



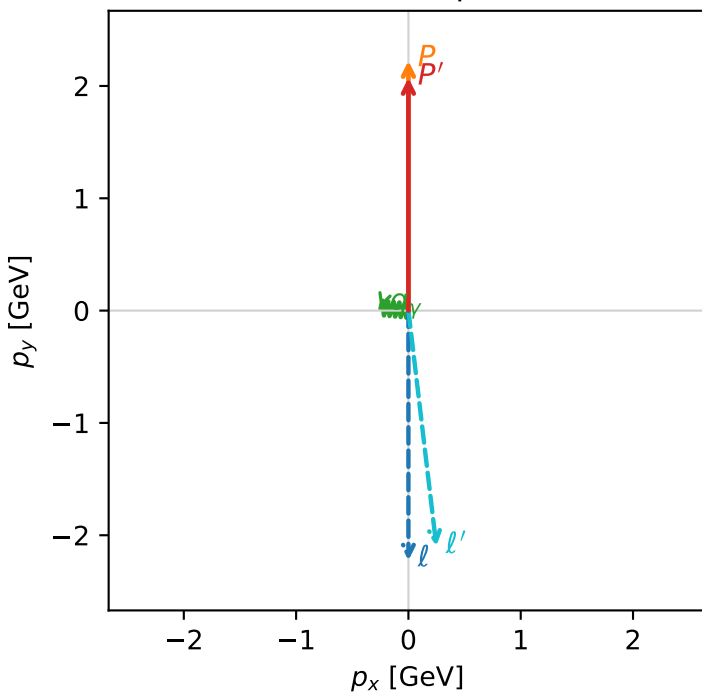
f3\_low\_egamma\_txtx\_region\_2  $F_3=0.166787$   
 region: low\_Egamma\_TxTx\_region\_2  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.07351$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=4.71239$ ,  $\phi_p=1.5708$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=3.04342$   
 $Q^2=0.0653998$ ,  $x_B=0.0593636$ ,  $t=-0.00364768$ ,  $W^2=1.91613$ ,  $y=0.0533864$   
 $\theta(\ell', \gamma)=102.388$  deg

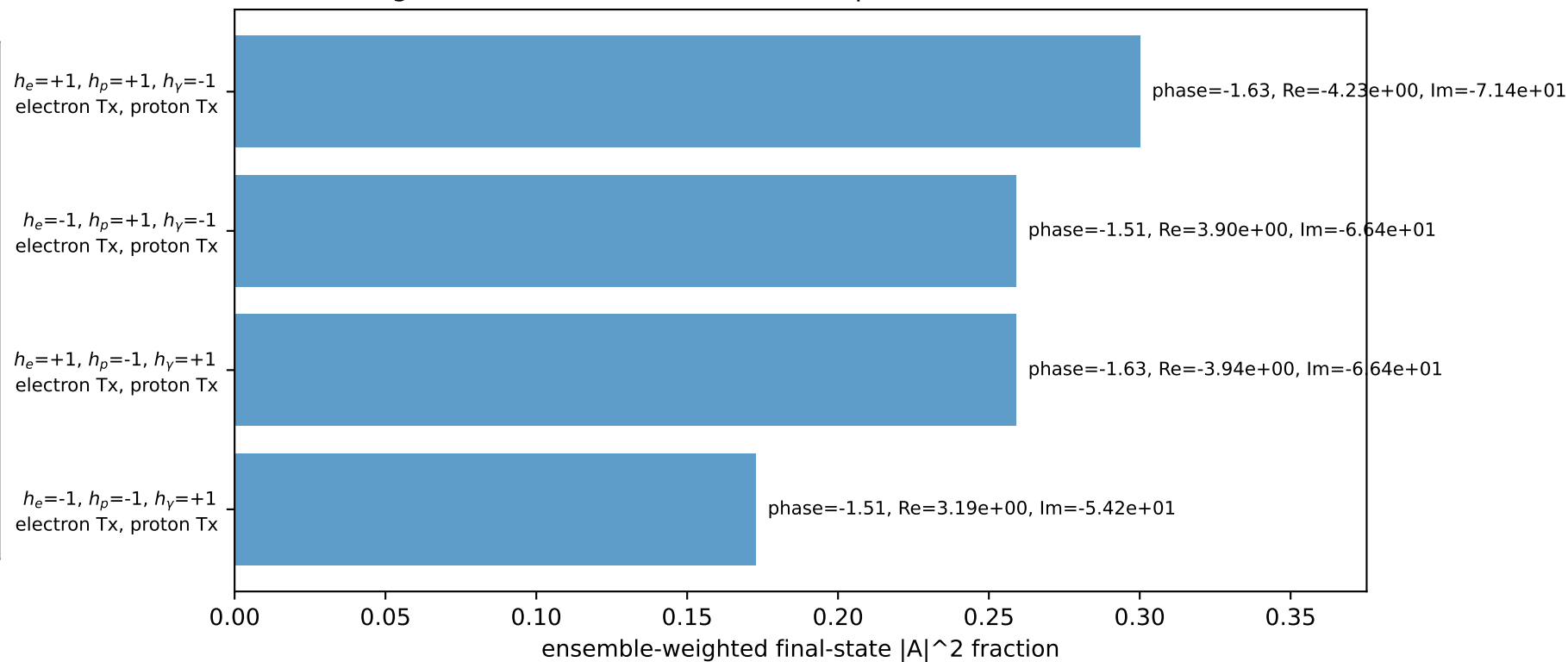
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$\ell$   $E= 2.2244$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 2.1127$   $p_x= 0.2488$   $p_y= -2.0980$   $p_z= 0.0000$   
 $P'$   $E= 2.2758$   $p_x= 0.0000$   $p_y= 2.0735$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= -0.2488$   $p_y= 0.0245$   $p_z= 0.0000$

Transverse plane

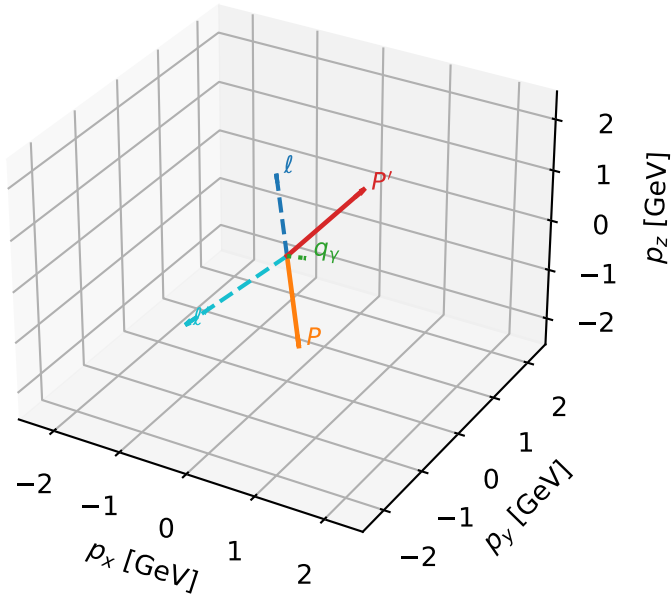


Leading initial-ensemble/final-state components (N=4, retained=99.1%)



# Tx massless lepton + Tx proton characteristic max $F_3$ configuration

3D momenta

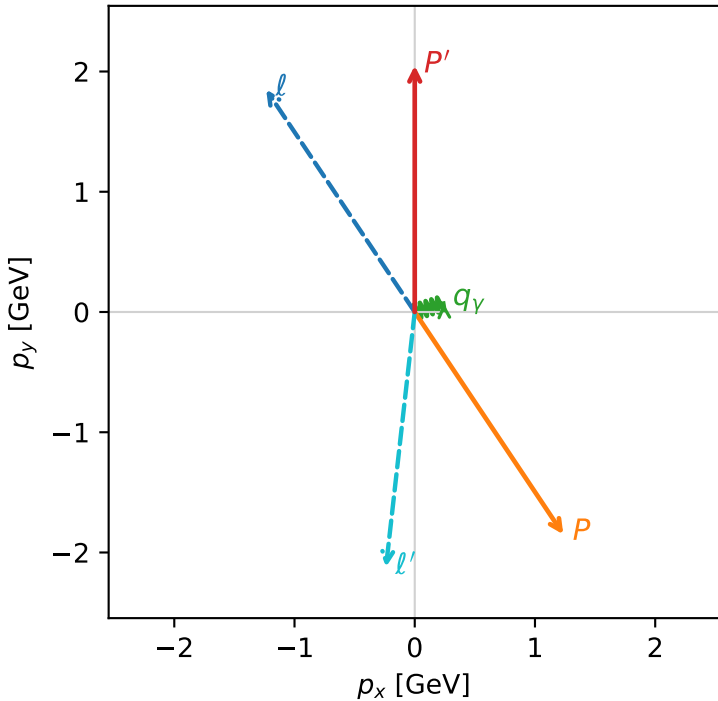


f3\_low\_egamma\_txtx\_region\_3  $F_3=0.140936$   
 region: low\_Egamma\_TxTx\_region\_3  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.049$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=2.15984$ ,  $\phi_p=5.30144$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=0.294524$   
 $Q^2=16.7549$ ,  $x_B=0.952836$ ,  $t=-16.7001$ ,  $W^2=1.7092$ ,  $y=0.852115$   
 $\theta(\ell', \gamma)=113.309$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -1.2358$   $p_y= 1.8495$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 1.2358$   $p_y= -1.8495$   $p_z= 0.0000$   
 $\ell'$   $E= 2.1350$   $p_x= -0.2392$   $p_y= -2.1216$   $p_z= 0.0000$   
 $P'$   $E= 2.2535$   $p_x= 0.0000$   $p_y= 2.0490$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.2392$   $p_y= 0.0726$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=98.6%)

